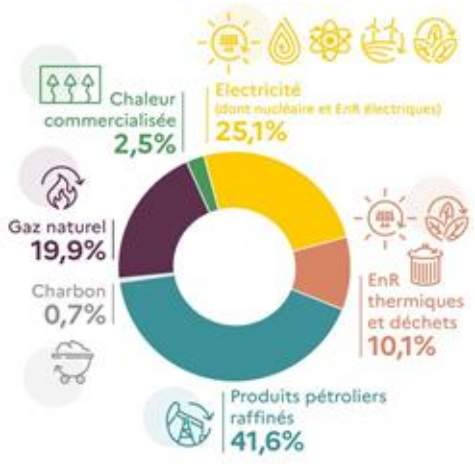
École des Ponts - 1ère année

Bases scientifiques pour la transition énergétique 1

Séance 6
Énergie du mouvement des fluides



Laurent BROCHARD

16 octobre 2026

École des Ponts ParisTech

Planning du module

- 11/09 – Énergie, ressources et climat : la nécessité d'une transition
- 18/09 – Rayonnement 1 (Amphi + TD)
- 25/09 – Rayonnement 2 (TD seulement)
- 02/10 – Énergie nucléaire 1 (Amphi + TD)
- 09/10 – Énergie nucléaire 2 (TD seulement)
- ***16/10 (matin) – Énergie du mouvement des fluides 1 (Amphi + TD)***
- 16/10 (après midi) – Énergie du mouvement des fluides 2 (TD seulement)
- 07/11 – Examen

Forms of energy

Kinetic energy	Potential energy
Mechanical: Motion (ordered kinetic energy)	**Mechanical**: Work (inter-molecular interactions)
Thermal (disordered kinetic energy)	
Electro-magnetic: Radiation (photon); Electrical (charged particles)	
	Chemical (intra-molecular interactions between electrons and nuclei)
	Nuclear (interactions between nucleons)

- Covered in classe prépa
- Lesson 2 & 3 - 18/09 & 25/09
- Lesson 4 & 5 - 02/10 & 09/10
- Covered in classe prépa
- **Lesson 6 & 7 - 16/10**
- Future lesson ?

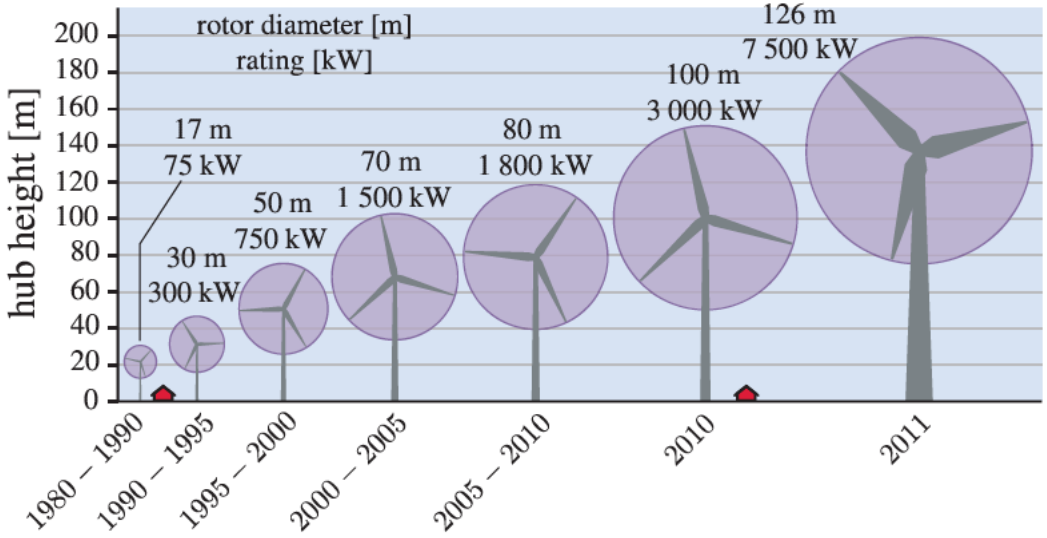
Exploitation of fluid motion: A long history





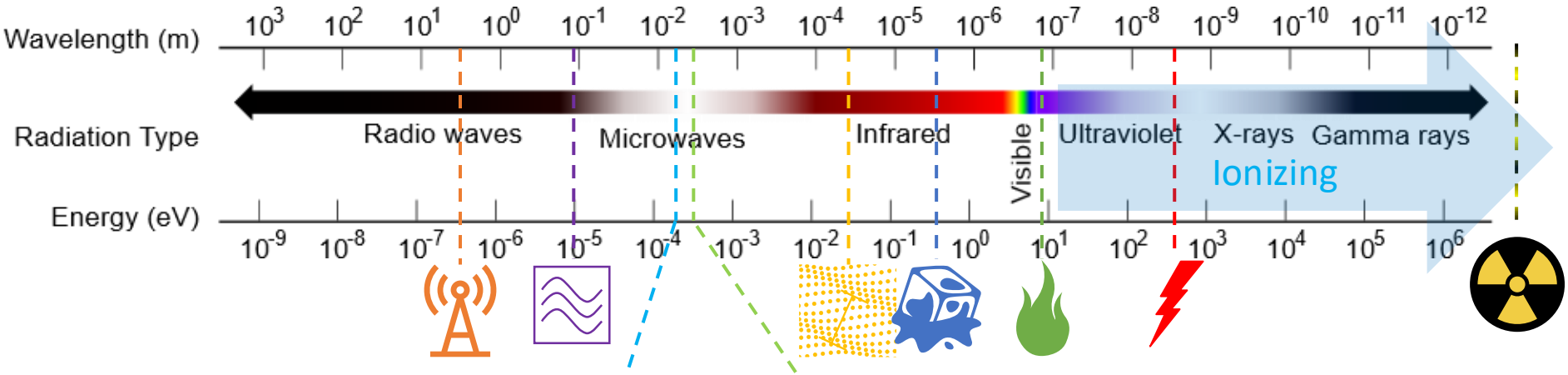


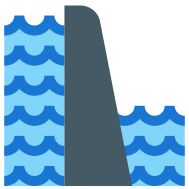




Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press

Scales of energy





Gravitation energy of a water molecule falling from a 100 m dam:
1.8×10^{-4} eV



Kinetic energy of an average air molecule associated with the collective motion of wind during a hurricane (50 m/s)
3.9×10^{-4} eV

Plan of the lesson

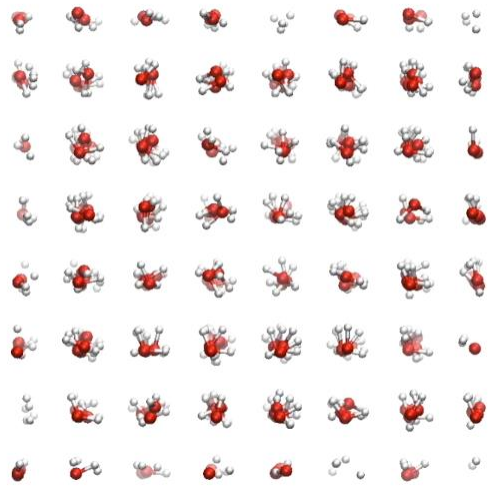
1. Fluid mechanics
2. Wind energy
3. Water-based energy

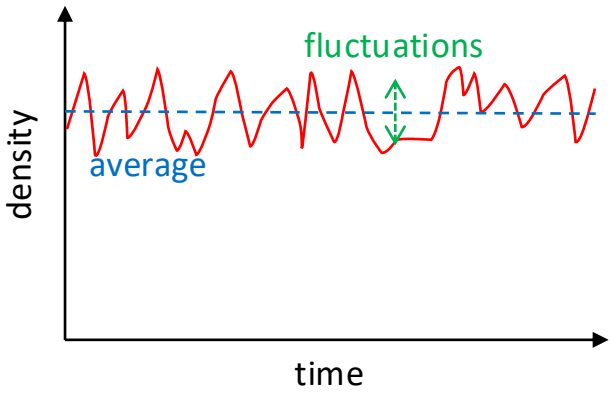
Plan of the lesson

1. Fluid mechanics
   - Recall of the basics
   - Working principle of an airfoil
2. Wind energy
3. Water-based energy

From molecules to continuum

- Nanoscale description: atoms and molecules





Statistical physics:

- Permanent thermal agitation
- Fluctuation decrease with system size in $N^{-1/2}$
- Only statistical average for macroscopic sizes ($N \sim 10^{23}$)

- Continuum description: thermodynamics
  - State parameters (e.g., (N, P, T))
  - Equilibrium = minimization of a thermodynamic potential (e.g., G(N, P, T))
  - No fluctuations, thermal agitation hidden in internal energy

Mechanical energy in the continuum description

1. Mechanical energy associated with deformation
   - Reminder: internal energy balance (volume change only)

$$dU = \underbrace{TdS}_{\text{heat}} - \underbrace{PdV}_{\text{work}}$$

   - Can be generalized to deformation in general $-PdV \rightarrow \underline{\underline{\sigma}} : d\underline{\underline{\varepsilon}}$

2. Mechanical energy associated with motion
   - Kinetic energy:

$$K = \sum_{i \in \text{atoms}} \frac{1}{2} m_i \mathbf{v}_i^2 = \frac{1}{2} \underbrace{m_{tot}}_{\sum_{i \in \text{atoms}} m_i} \mathbf{v}_{\text{macro}}^2 + \sum_{i \in \text{atoms}} \frac{1}{2} m_i \left(\mathbf{v}_i^2 - \mathbf{v}_{\text{macro}}^2 \right)$$

energy of the macroscopic motion
- "ordered kinetic energy"
- Not part of internal energy
- Fully recoverable

$\ll$ At ambient T

energy of the thermal agitation:
- "disordered kinetic energy"
- Part of internal energy
- Recovery limited by second principle

Continuum properties

- Local state in a continuum: intensive properties

  N, V, U → N/V, U/V, V/N, U/N etc…

  - Mass density: $\rho(\mathbf{x}, t)$ = (N/V).m
  - Internal energy per unit mass: $u(\mathbf{x}, t)$ = (U/N)/m

- Conjugated quantities (already intensive)
  - Stress: $\underline{\underline{\sigma}}(\mathbf{x}, t) = -P\underline{\underline{I}} + \underline{\underline{\tau}}$ (pressure and deviatoric stress)
  - Temperature: $T(\mathbf{x}, t)$

- Macroscopic motion
  - Velocity: $\mathbf{v}(\mathbf{x}, t) = d\mathbf{x}/dt$
  - Kinetic energy density: $\rho(\mathbf{x}, t)e_{kin}(\mathbf{x}, t) = ½\ \rho|\mathbf{v}|^2$
  - Total energy per unit mass: $e(\mathbf{x}, t) = u + e_{kin}$

Flux

- Mass flux: $\mathbf{J}_\rho = \rho\,\mathbf{v}$
- Momentum flux (X-axis): $\mathbf{J}_{px} = \rho\, v_x\, \mathbf{v}$
- Energy flux: $\mathbf{J}_e = \rho\, e\, \mathbf{v} = \rho\, u\, \mathbf{v} + \underbrace{\rho\, e_{kin}\, \mathbf{v}}_{\mathbf{J}_{e_{kin}}}$ Macroscopic kinetic energy flux
- Power density: $\mathcal{P} = |\mathbf{J}_{e_{kin}}(\mathbf{x},t)| = \frac{1}{2}\rho(\mathbf{x},t)\,|\mathbf{v}(\mathbf{x},t)|^3$

energy of motion crossing a surface orthogonal
to the flow per unit time and area

→Energy of motion available

❿ ρ_{air} = 1.3 kg/m^3

❿ ρ_{eau} = 1000 kg/m^3

Wind at 30 km/h → $\mathcal{P}$ = 1 740 W/m^2
≈ Water stream at 5 km/h
> Solar radiation above the atmosphere (1 380 W/m^2)

What is conserved ?

- 1st principle: total energy in absence of external sources
  - Internal energy u
  - Macroscopic kinetic energy e_{kin}

→Conservation of energy

  - Rq: mass is energy (relativity)

but for non-relativistic systems mass changes are negligible

→Conservation of mass

- In absence of external forces:
  - Translational symmetry → conservation of linear momentum
  - Rotational symmetry → conservation of angular momentum

→Conservation of momentum (vectorial)

Conservation of mass, momentum and energy

- Mass: $A(\mathbf{x}, t) = \rho(\mathbf{x}, t)$ and $S_A(\mathbf{x}, t) = 0$

$$\frac{\partial \rho}{\partial t} + \nabla.(\rho \mathbf{v}) = 0$$ Continuity equation

General form

$$\frac{\partial A}{\partial t} + \nabla.(A\mathbf{v}) = S_A$$

flux ↑ source ↑

- Momentum
  - Linear momentum: $A(\mathbf{x}, t) = \rho v_i$ and $S_A(\mathbf{x}, t)$ = stress on boundaries + body force (gravity)

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla.(\rho \mathbf{v} \otimes \mathbf{v}) = \nabla.\underline{\underline{\sigma}} + \rho \mathbf{g}$$ Navier-Stokes equation

outer product $\mathbf{u} \otimes \mathbf{v} = \mathbf{u}.\mathbf{v}^T$

  - Angular momentum: $A(\mathbf{x}, t) = \rho\,(\mathbf{x} \times \mathbf{v})_i$ and $S_A(\mathbf{x}, t)$ = moment of gravity and stress

→ Symmetry of stress tensor

- Energy: $A(\mathbf{x}, t) = \rho(\mathbf{x}, t)e(\mathbf{x}, t)$ and $S_A(\mathbf{x}, t)$ = work and heat exchanges at boundaries + work of body force (gravity)

$$\frac{\partial (\rho e)}{\partial t} + \nabla.(\rho e \mathbf{v}) = \nabla.(\underline{\underline{\sigma}}.\mathbf{v}) + \rho \mathbf{g}.\mathbf{v} + \nabla.\mathbf{q}$$ Energy balance equation

Complete problem

- General problem

$$\begin{cases} \frac{\partial \rho}{\partial t} + \nabla.(\rho \mathbf{v}) = 0 \\ \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}.\nabla \mathbf{v} = \frac{\nabla.\underline{\underline{\sigma}}}{\rho} + \mathbf{g} \\ \frac{\partial (\rho e)}{\partial t} + \nabla.(\rho e \mathbf{v}) = \nabla.(\underline{\underline{\sigma}}.\mathbf{v}) + \rho \mathbf{g}.\mathbf{v} + \nabla.\mathbf{q} \end{cases}$$

- For completeness, one needs in addition
  - a behavior law: $(u, \underline{\underline{\sigma}}) = f(\rho, T, \mathbf{v})$

    Ex: Diatomic ideal gas $p = \frac{\rho R T}{M}$ and $u = \frac{5}{2}\frac{RT}{M}$ and no shear
  - a heat flux law: $\mathbf{q} = g(\rho, T, \mathbf{v})$

    Ex: Fourier's law $\mathbf{q} = -\kappa \nabla T$

- →5 unknowns (1 ρ, 3 **v**, 1 T) for 5 equations (1 mass, 3 momentum, 1 energy)

Simplifications for air and water flows

- Common simplifying assumptions
  1. Incompressible: $\rho = \rho_0$
  2. Adiabatic: $\mathbf{q} = 0$
  3. Newtonian fluid: $\underline{\underline{\sigma}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$ (pressure; deviatoric stress) with $\underline{\underline{\tau}} = \mu\left(\nabla\mathbf{v} + \nabla\mathbf{v}^T\right)$ (dynamic viscosity)

$$\nabla.\mathbf{v} = 0$$ Continuity

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}.\nabla\mathbf{v} = -\frac{\nabla p}{\rho} + \frac{\mu}{\rho}\Delta\mathbf{v} + \mathbf{g}$$ Navier-Stokes ($\nabla.\underline{\underline{\sigma}}$)

$$\frac{\partial e}{\partial t} + \mathbf{v}.\nabla e = -\frac{\mathbf{v}.\nabla p}{\rho_0} + \frac{\nabla.\left(\underline{\underline{\tau}}.\mathbf{v}\right)}{\rho_0} + \mathbf{g}.\mathbf{v}$$ Energy balance

  4. Stationary:
     excludes turbulent flows
  5. Ideal fluid: non viscous ($\mu = 0$)
     valid for small velocity gradient only, cannot be true in boundary layer

Bernoulli's principle

- Hypothesis:
  - Ideal fluid (no viscosity + adiabatic = isentropic)
  - Stationary

- Conservation of mass + conservation of energy

$$u + \frac{\mathbf{v}^2}{2} + \frac{p}{\rho} + gz \quad \text{constant throughout the flow}$$

→ Slowing down a fluid flow increases p and u (i.e., T)
→ Accelerating a fluid flow decreases p and u (i.e., T)

- Case of wind (diatomic ideal gas) at $|\mathbf{v}| = 50$ km/h
  - Max change of pressure for $\Delta u = 0$ and $\Delta z = 0$: $\Delta p \approx 1.2$ hPa $\ll 1$ atm

→Relative pressure change $\Delta p/p \sim 10^{-3} \Rightarrow$ same for density $\Delta\rho/\rho \sim 10^{-3}$

  - Max change of temperature for $\Delta p = 0$ and $\Delta z = 0$: $\Delta T = \Delta u/C_p \approx 0.1\text{K} \ll 300\text{K}$

→Relative temperature change $\Delta T/T \sim 3.10^{-5} \ll \Delta p/p$

Temperature changes can be neglected & fluid can be assumed incompressible

Bernoulli's principle

- Hypothesis:
  - Ideal fluid (no viscosity + adiabatic = isentropic)
  - Stationary
  - Incompressible: $\rho = \rho_0$ and no work $u = u_0$ ↓ no heat
- Conservation of mass + conservation of energy

$$\frac{\mathbf{v}^2}{2} + \frac{p}{\rho} + gz \quad \text{constant throughout the flow}$$

→ Slowing down a fluid flow increases *p*

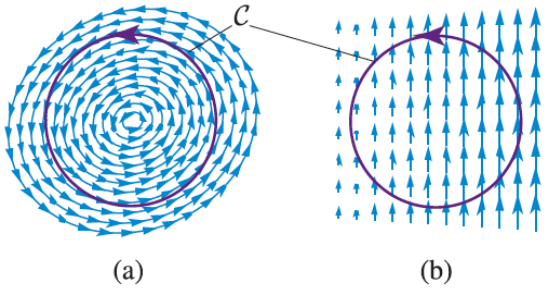
→ Accelerating a fluid flow decreases *p*

- 'dynamic' pressure $p_{dyn} = \frac{\rho \mathbf{v}^2}{2}$: excess pressure if fluid motion stopped

Vorticity and circulation

- Definition:

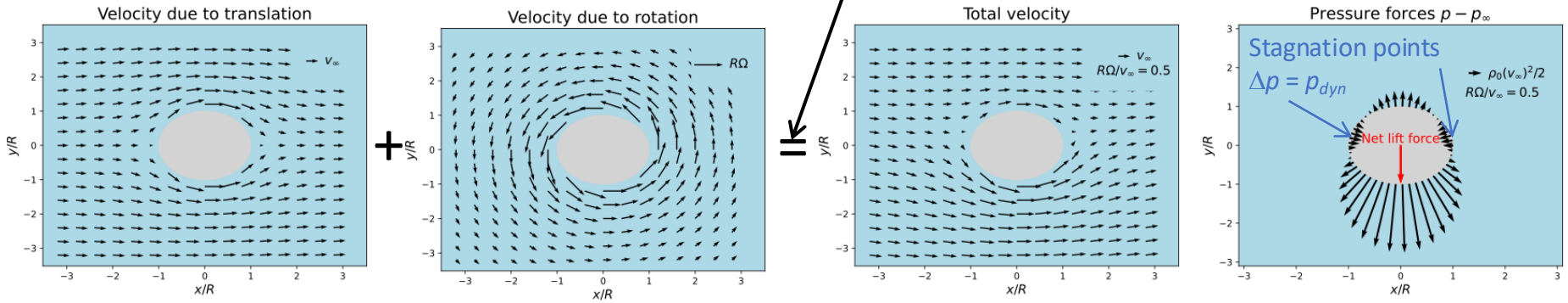
$$\underbrace{\Gamma(\mathcal{C})}_{\text{circulation}} = \oint_{\mathcal{C}} \mathbf{v}.d\mathbf{x} = \int_S \underbrace{(\nabla \times \mathbf{v})}_{\omega \text{ vorticity}}.d\mathbf{S}$$



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- Kelvin's circulation theorem: the vorticity of a packet of ideal fluid is constant ⇒ in particular, if irrotational ($\omega = 0$) remains irrotational.
- Real fluids: viscosity eventually dissipates vorticity
- Around an object, the circulation can be nonzero even if fluid irrotational ($\omega = 0$)
- Example of application: magnus effect

Superposition because fluid mech. linear for steady flow of ideal incompressible fluid



Irrotational
No circulation

Irrotational
Circulation

Irrotational
Circulation

Induced pressure change:
net lift force

Kutta-Zhukovskii theorem

- Hypothesis: stationary flow of an incompressible ideal fluid around an object (infinite along the axis with same cross-section)
- Demonstration: see notes (uses Bernoulli + incompressible + irrotational)
- Lift per unit length

axis of the object

direction of the flow

$$\mathbf{f}_l = -\rho_0 v_\infty \Gamma(\mathcal{C})\, \mathbf{a} \times \mathbf{u}$$

lift due to circulation

free-stream velocity

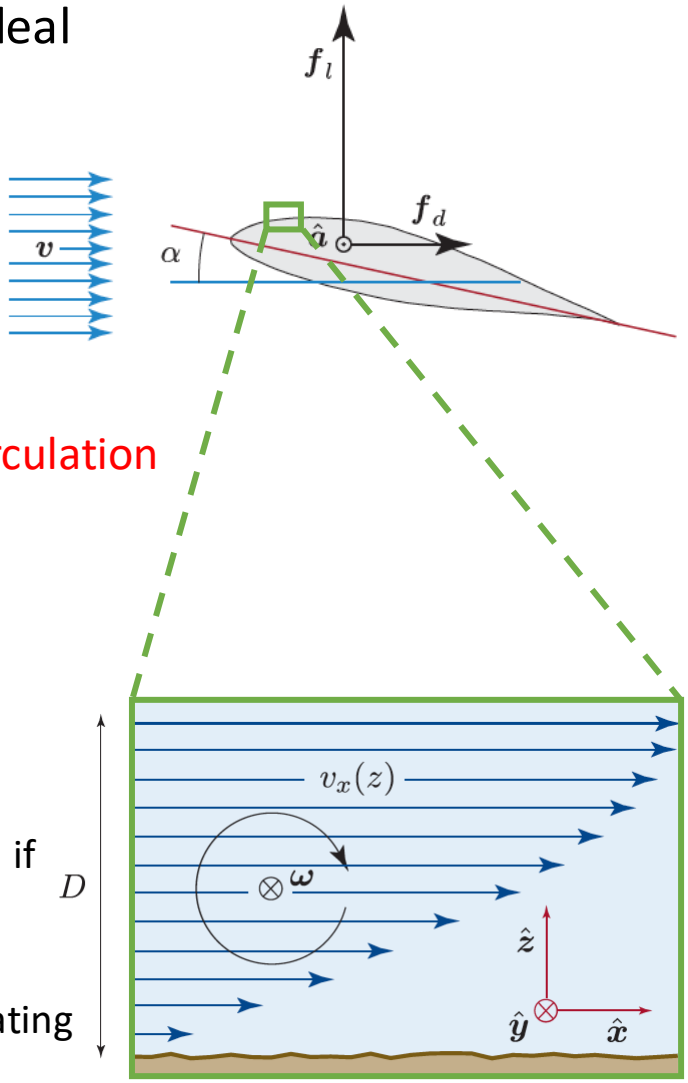
any closed curve around the object oriented counterclockwise

- Drag per unit length

$$\mathbf{f}_d = 0$$

drag due to viscosity

- True fluids are not ideal (except He superfluid…)
  - Always a boundary layer at the surface of the object (even if small) → net drag force in the direction of the flow
  - In the layer: vorticity and Kutta-Zhukovskii not applicable
  - Boundary layer also essential for the formation of a circulating flow around an airfoil from an initially irrotational flow



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Role of viscosity

- Dimensionless formulation of incompressible Navier-Stokes

$$\frac{\partial \mathbf{v}^*}{\partial t^*} + \mathbf{v}^*.\nabla_{\mathbf{x}^*}\mathbf{v}^* = -\nabla_{\mathbf{x}^*}p^* + \frac{1}{\mathcal{R}e}\Delta_{\mathbf{x}^*}\mathbf{v}^* + \frac{1}{\mathcal{F}r^2}\mathbf{e}_z \text{ with } \begin{cases} \mathbf{x}^* = \frac{\mathbf{x}}{L} \\ \mathbf{v}^* = \frac{\mathbf{v}}{v_\infty} \\ p^* = \frac{p}{\rho_0 v_\infty^2} \\ t^* = \frac{v_\infty}{L}t \end{cases}$$

Reynolds — Dynamic pressure

$$\mathcal{R}e = \frac{\rho_0 v_\infty^2}{\mu v_\infty / L}$$

Characteristic shear stress

Froude

$$\mathcal{F}r = \frac{v_\infty}{\sqrt{gL}}$$

v_∞ : free stream velocity
ρ_0 : characteristic density
L : characteristic length

→Useful for experiments on mock-ups (e.g., wind-tunnels)

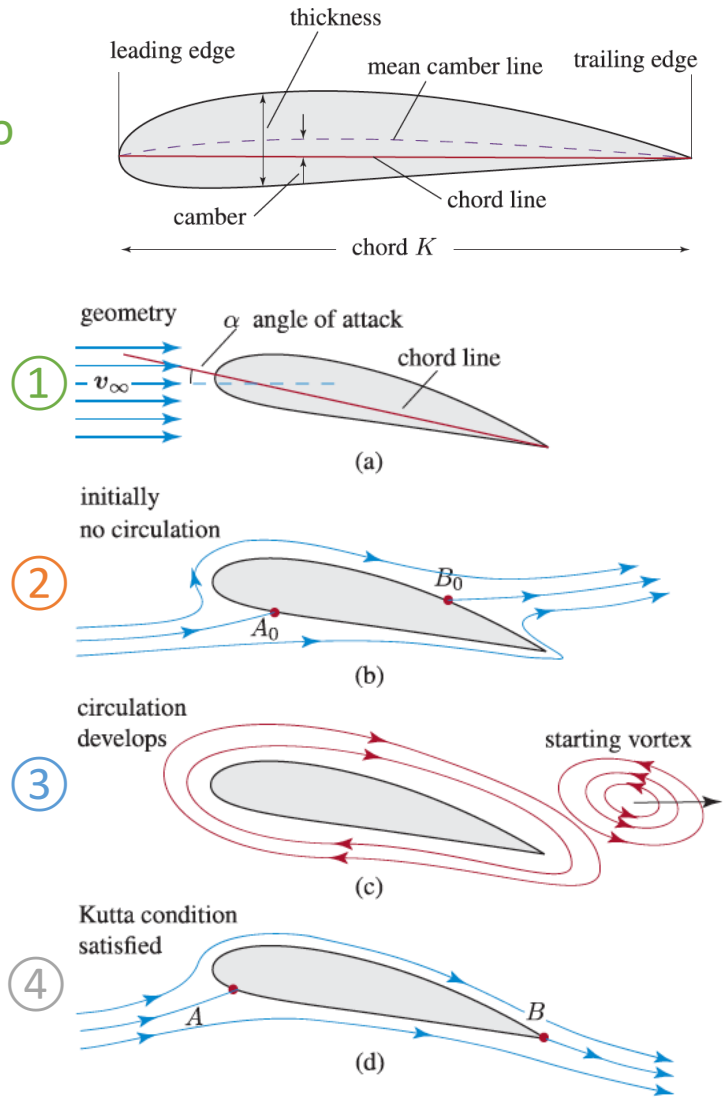
- Viscosity dominates if $\mathcal{R}e \lesssim 10^3$ - 10^4
  - For pipes (diameter L): laminar flow if $\mathcal{R}e \lesssim 2300$, turbulent flow if $\mathcal{R}e \gtrsim 4000$
  - For an airfoil (chord length L = 1m) in a v_∞ = 50 km/h wind: $\mathcal{R}e \sim 10^6$
  - For an hydrofoil (chord length L = 20 cm) in a v_∞ = 1.5 km/h stream: $\mathcal{R}e \sim 10^5$

  →Viscosity negligible except for the boundary layer

Circulation around an airfoil: Kutta condition

- How circulation develops?
  1. Airfoil begins to move with an angle of attack α. The flow is irrotational (Kelvin), no circulation, no lift.
  2. Two stagnation points appear (A_0 & B_0), that would persist indefinitely for an ideal fluid, with infinite velocity at the sharp trailing edge.
  3. High velocities between B_0 and the trailing edge induce non negligible viscous forces in the boundary layer, initiating vorticity and a clockwise circulation around the airfoil.
  4. The vortex moves away in the wake leaving an irrotational circulation around the airfoil, with the rear stagnation point (B) shifted to the sharp trailing edge.

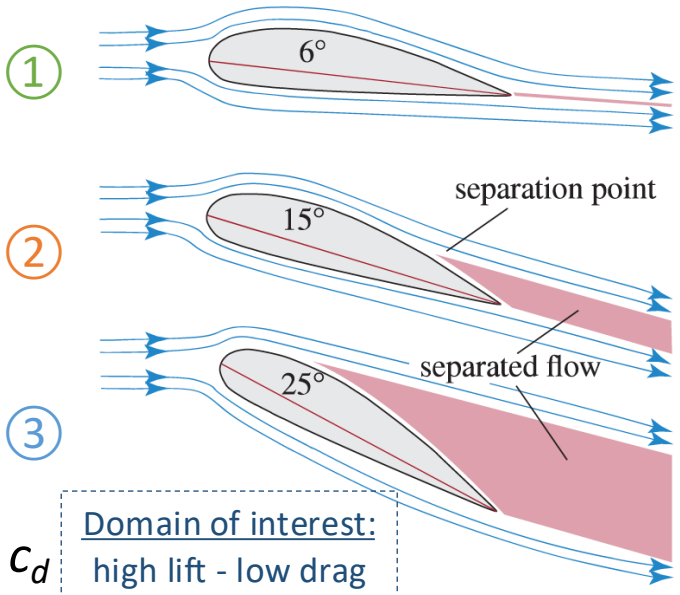
- <u>Kutta condition:</u> When an airfoil **with a sharp trailing edge** is immersed in a uniform fluid flow, viscous forces in the boundary layer establish a circulation of exactly the magnitude necessary to keep the rear stagnation point at the trailing edge of the airfoil.



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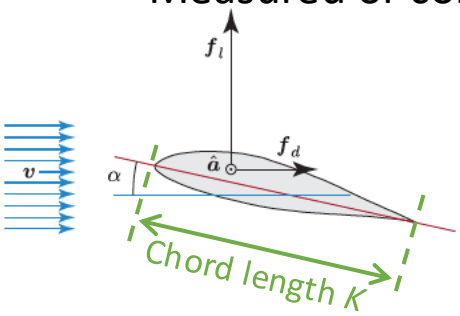
Circulation around an airfoil: Kutta condition

- Kutta condition fails as α increases:
  1. Under Kutta condition at small α: very narrow region of turbulent wake directly behind its trailing edge
  2. At larger α, the steady flow separates from the top of the airfoil leaving behind an enlarged region of turbulence.
  3. Lift start to decrease as the turbulent region becomes to large: the airfoils stalls.



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- Characterization with lift and drag coefficients c_l and c_d
  - Dimensionless
  - Characterized as a function of α
  - Measured or computed by CFD



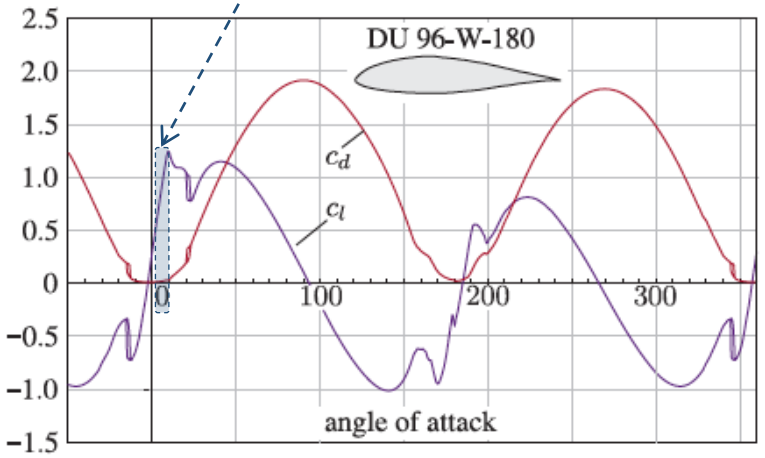
Chord length

$$\mathbf{f}_l = c_l \frac{\rho \mathbf{v}^2}{2} K \mathbf{a} \times \mathbf{u}$$

$$\mathbf{f}_d = c_d \frac{\rho \mathbf{v}^2}{2} K \mathbf{u}$$

p_{dyn}

Domain of interest:
high lift - low drag



Plan of the lesson

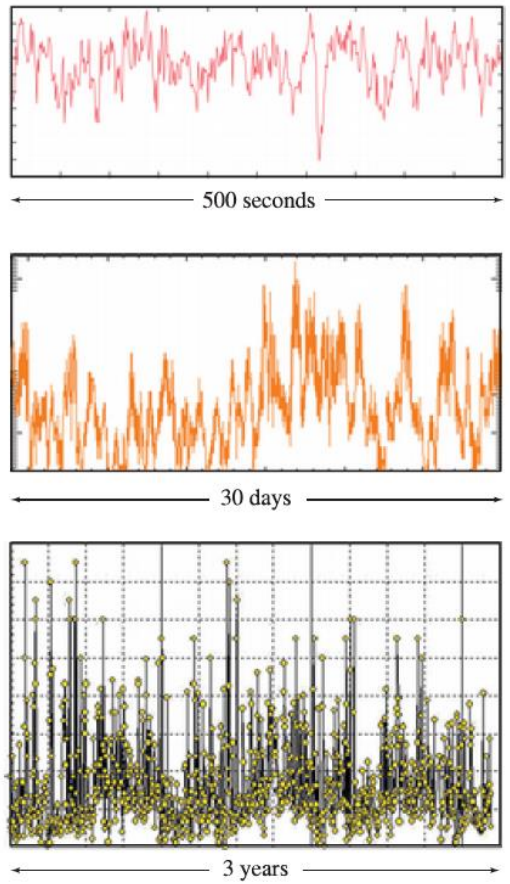
1. Fluid mechanics

2. Wind energy
   - Wind resource
   - Betz limit
   - Horizontal axis wind turbines
   - Alternative designs

3. Water-based energy

A highly variable resource

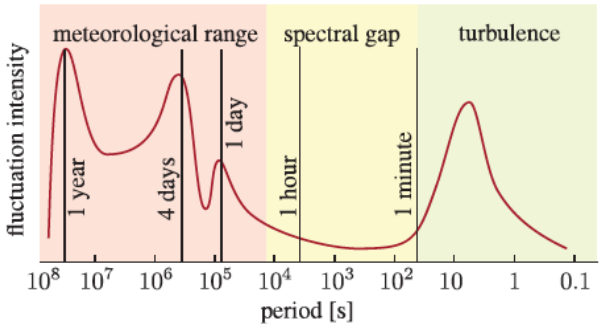
Time series of wind speed (three different time scales)



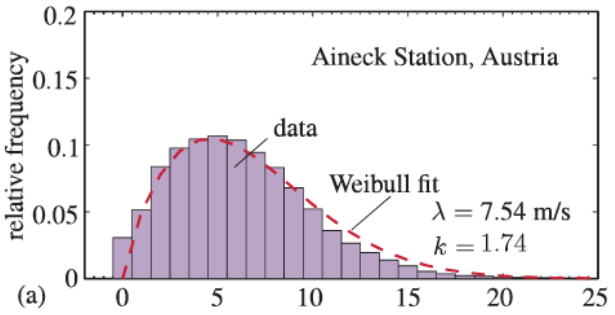
Fourier transform

Statistical distribution

Wind fluctuations spectrum



Wind speed frequency



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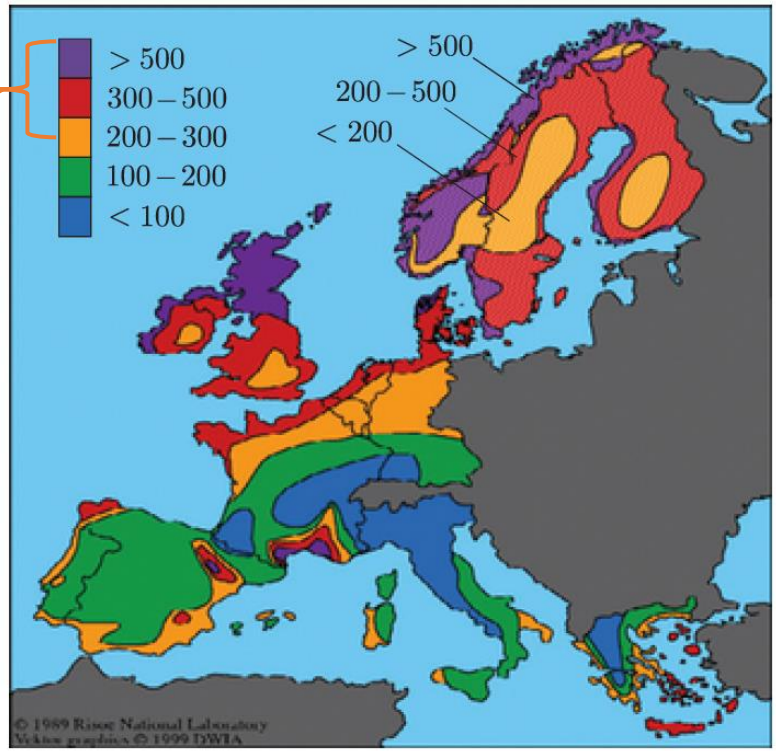
A widely available resource

- Resource available:
  - 1 to 100 TW (global need 20 TW)
  - Best locations: coast, open ocean, hills, ridges…
- But:
  - Sometimes far from consumers (e.g., offshore)
  - Disturbed wind in the wake of a turbine → available power does not scale with the area
  - Intermittency
- Currently: 0.8 TW installed, with capacity factor between 20% and 40%

Most suitable for utility scale wind power production

Annual average wind power atlas (Western Europe)

Wind power in W/m² (50 m over the ground)



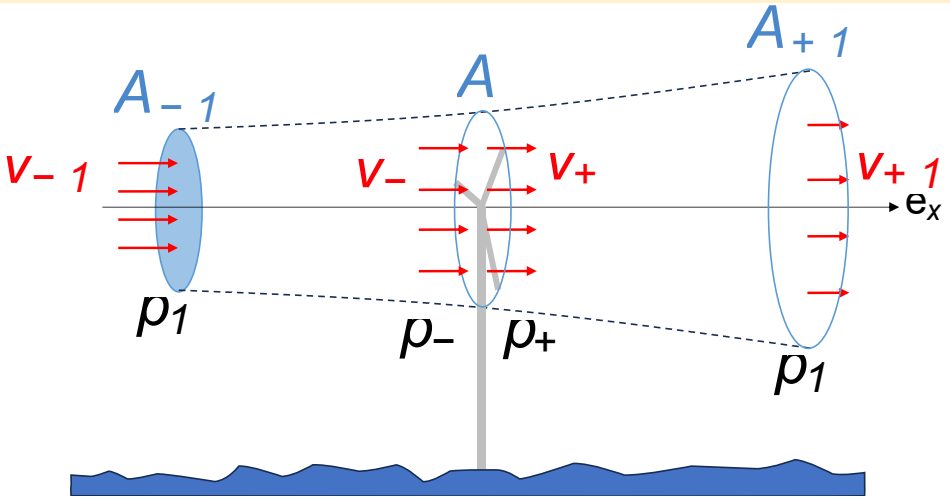
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Betz limit

- Hypothesis:
  - Incompressible ideal fluid
  - Steady flow
  - Axial momentum analysis
- Efficiency:

$$C_p = \frac{\mathcal{P}}{\frac{1}{2}\rho_0 m \left(v_{-\infty}\right)^3 A} = \frac{16}{27} \approx 59\%$$

Total kinetic power of an undisturbed wind flowing through the section of the wind turbine



- Demonstration: from conservation of mass, axial momentum and Bernoulli →See exercise in small class
- Note: Qualitatively different from Carnot or Shockley-Queisser
  - Carnot / S-Q = absolute upper bound on the energy that be usefully extracted

heat to work

work to work

  - Betz = the energy not extracted still available for further turbines

Betz limit with wake rotation

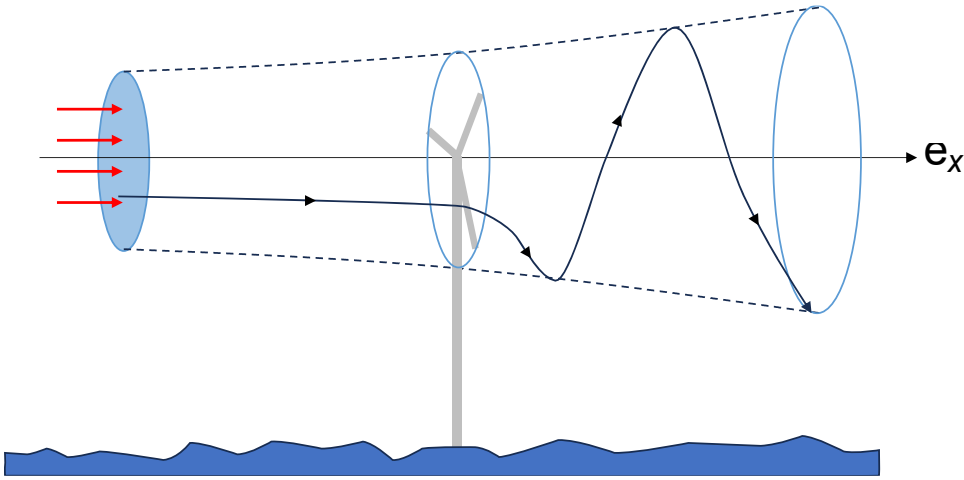
- Hypothesis:
  - Incompressible ideal fluid
  - Steady flow
  - Axial & angular momentum analysis
- Demonstration: →See lecture notes
- Efficiency:
  - Depends on tip-speed ratio

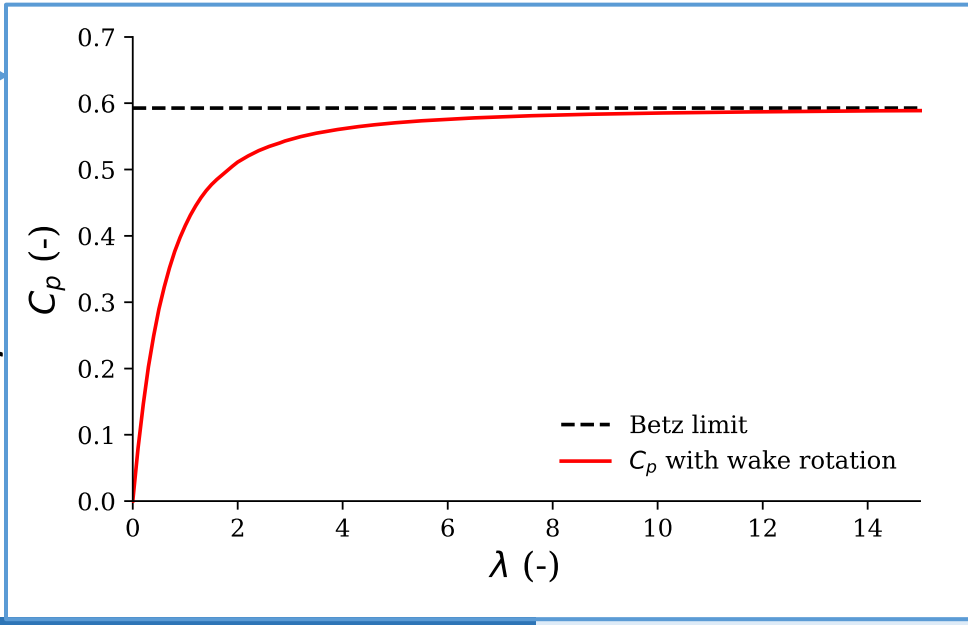
$$\lambda = \frac{\Omega R}{v_{-\infty}}$$

Blade tip rotation velocity

Wind velocity

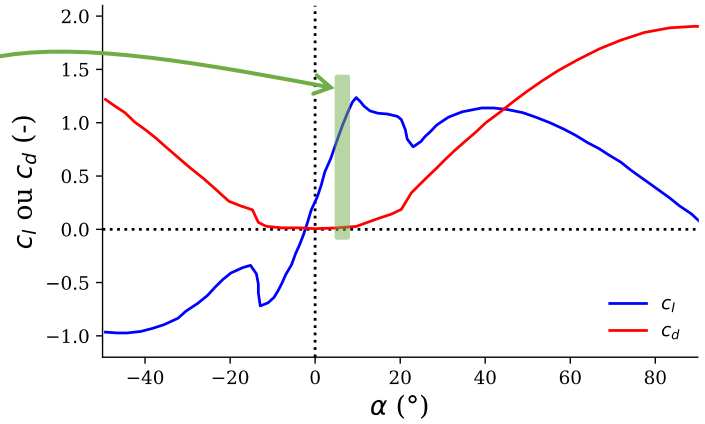
  - Betz limit for $\lambda >> 1$, but much lower at slow rotation because more torque needed for a given power and wake rotation originates from torque

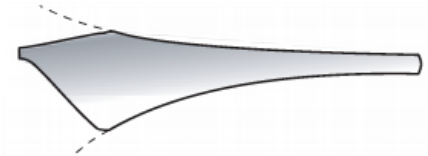


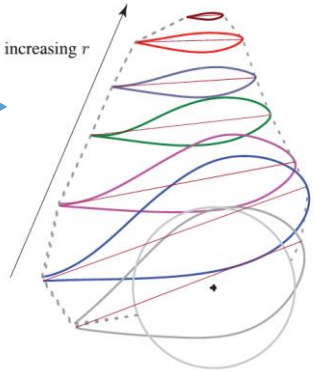


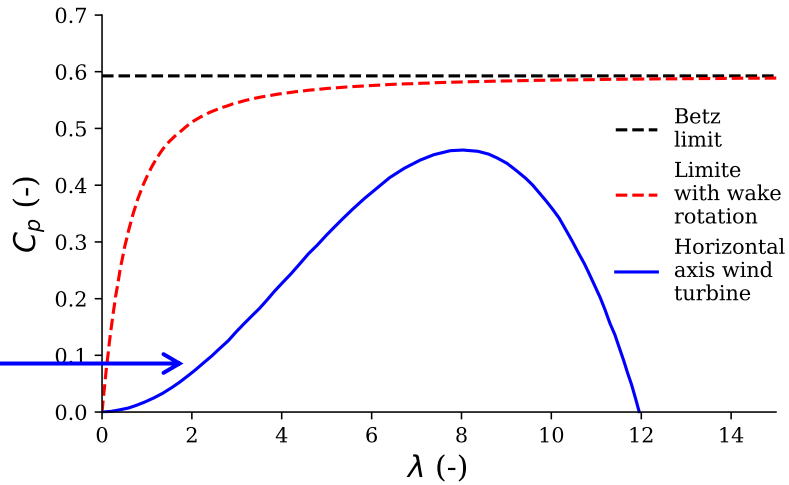
Design of a wind turbine

- Optimization problem
  1. Optimal angle of attack
     - High lift
     - low drag
  2. Variable apparent wind velocity along the blade depending on tip-speed ratio λ
  3. Design choices
     - Chord length
     - Twist angle
     - Number of blades









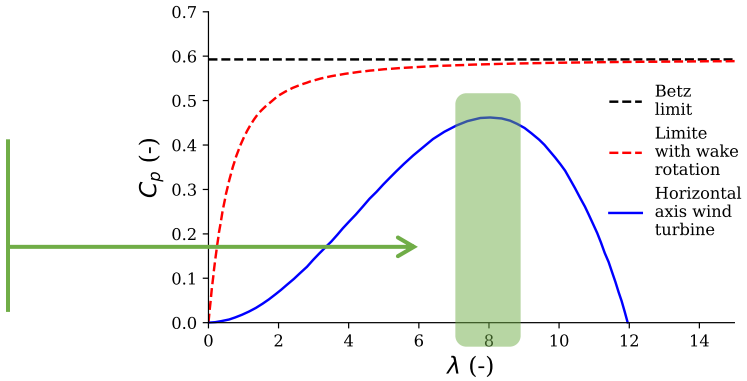
- Optimum only <u>for a given λ</u>

See exercise in small class

Operate a wind turbine

- Rotation velocity

→ Adapted to wind velocity to maintain constant tip-speed ratio λ



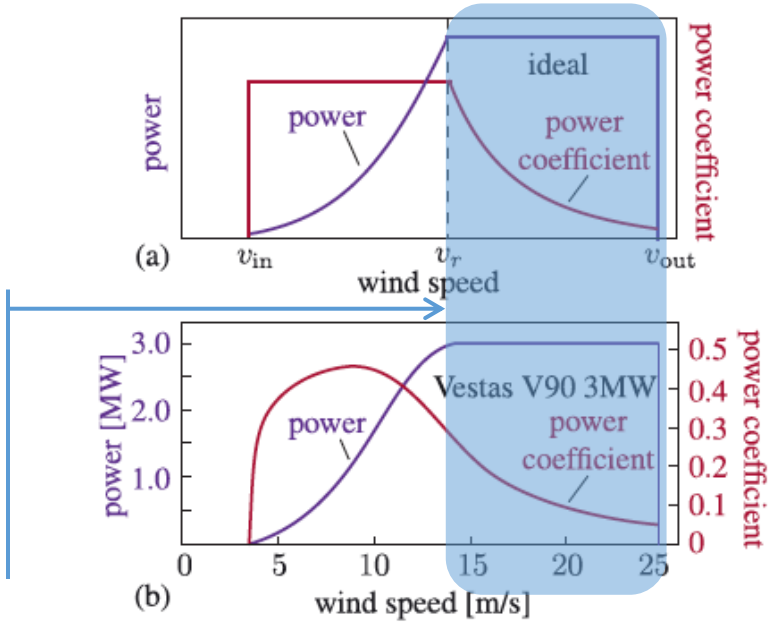
- Wind direction

→ active yaw control (motor)

- Too high winds

→ pitch-regulated : blades rotated (lower the lift, increase the drag)

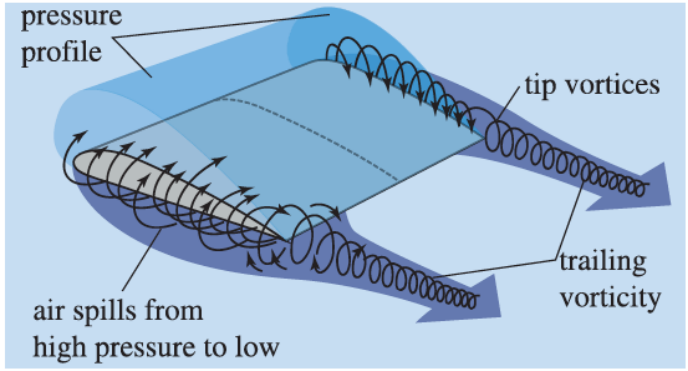
→ stall-regulated : blade shaped to stall progressively under too high speed

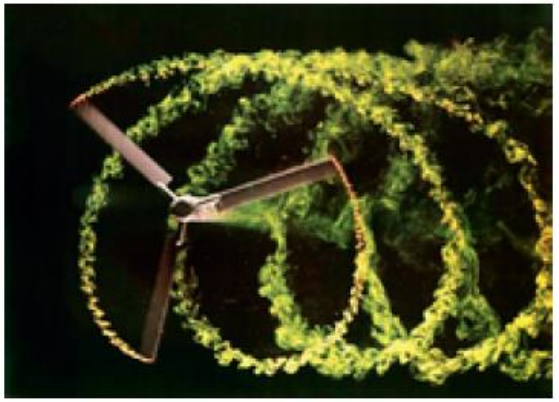


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Efficiency issues

- Blade tip vortices





- Wind shadow in the wake of a turbine

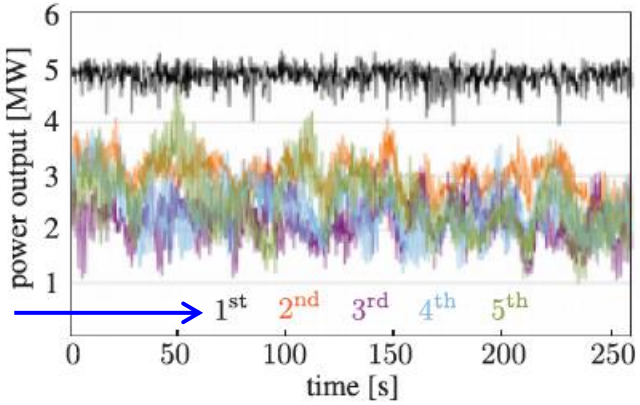
~2-5 W/m^2 (on average)

→Spacing of ~ 10 R vs. area of πR^2 ⇒ power density of a farm independent of R

→2nd order: larger R ⇒ larger wind, more stable, less materials

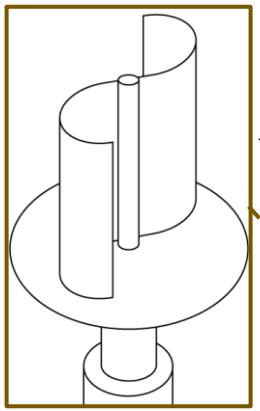


5 successive turbines separated by 3 diameters



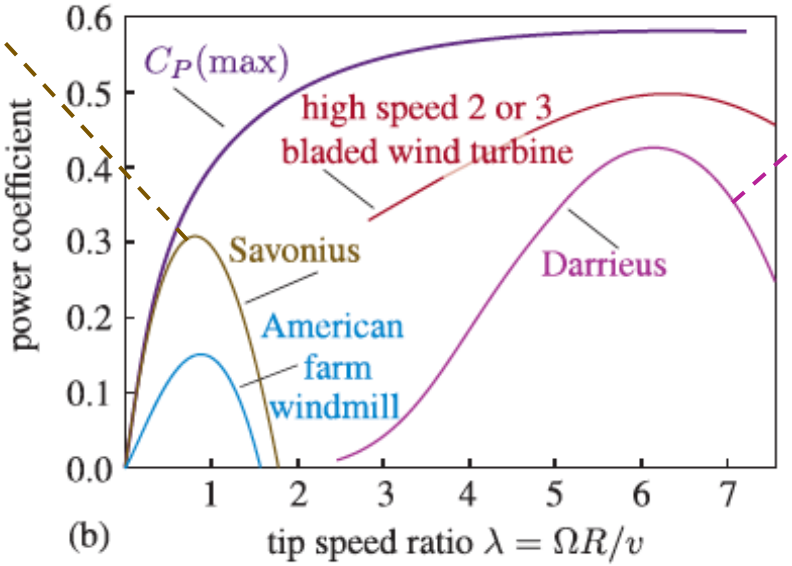
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Other types of wind turbines



Vertical axis





See exercise in small class

Exploiting drag force

ancestral panemone



In water

Tidal stream generator



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Plan of the lesson

1. Fluid mechanics

2. Wind energy

3. Water-based energy
   - Hydropower
   - Waves
   - Tides and currents
   - Ocean thermal energy

Hydro : moderate resource but highly concentrated

- Most developed renewable electric power generation

→430 GW on average ≈ 1/6th of global electricity

- Estimated global potential : 1 600 GW
  vs. wind : 100 000 GW
  vs. global need : 20 000 GW
- Much more concentrated than wind (x 10^3)
  - Easy to harvest

  → Long history (> 2000 yrs)
- Two main types of hydropower
  - Reservoir dam
  - Run-of-river plant
- Renewable & <u>flexible</u>
  - On-demand,
  - Reservoir dam can be used for energy storage (pumping)



Wikimedia commons (CC BY-SA 3.0)



U.S. Army Corps of Engineers (Public Domain)

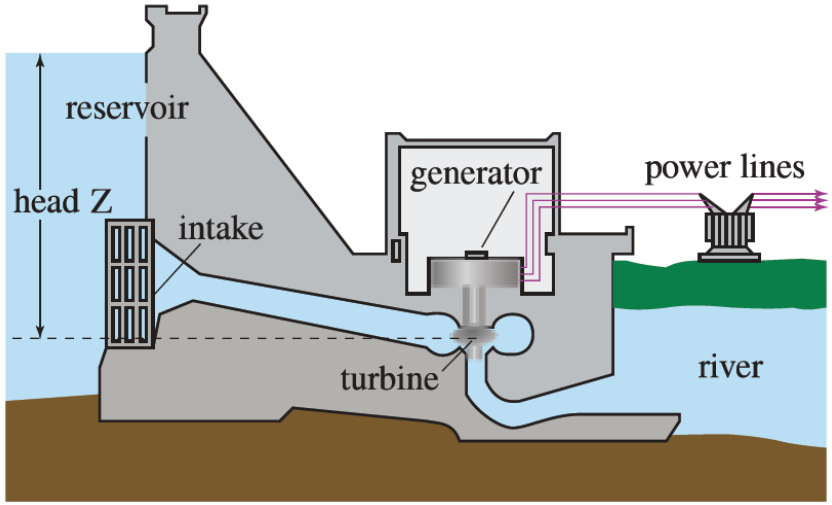


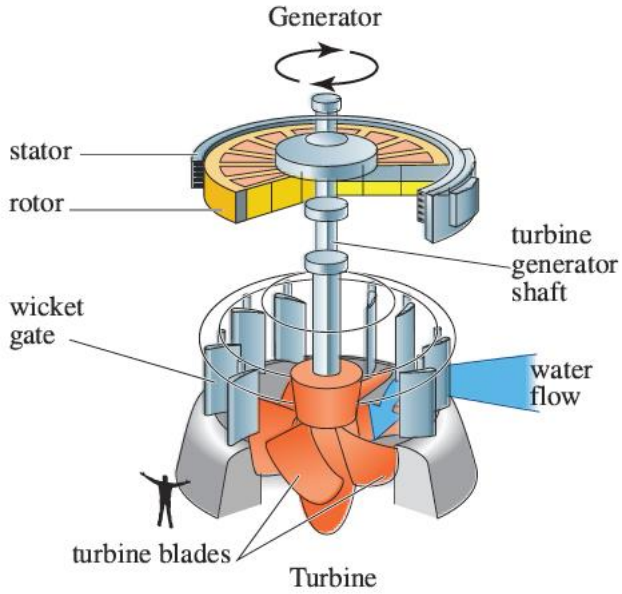
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Physics of hydropower

- Water gravitational energy $\rho g Z$ recovered by channeling water through a turbine that converts the hydrostatic pressure into rotation work/electricity
- Betz limit does not apply
  - Flow constrained to go through the turbine (no by-pass possible)

  →Can reach 100% efficiency





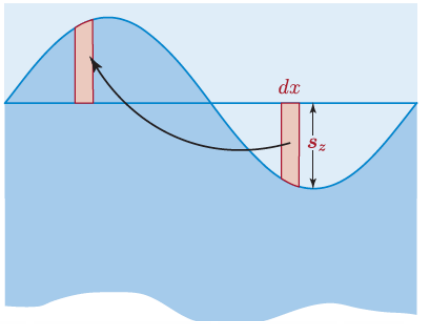
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Wave power

- Energy of waves
  - Surface energy density $\frac{E}{A} = \frac{1}{2}\rho g h$ travelling at velocity $v_g = \frac{g}{2\omega}$

Waves amplitude (height)

Angular frequency of the wave



  → Linear power density $\frac{\mathcal{P}}{L} = \frac{\rho g^2 h^2}{4\omega}$

  - Typically : 40 kW/m (1 m wave, 10 s period)

  → Substantial (e.g., av. Solar : 0.15 kW/m²)

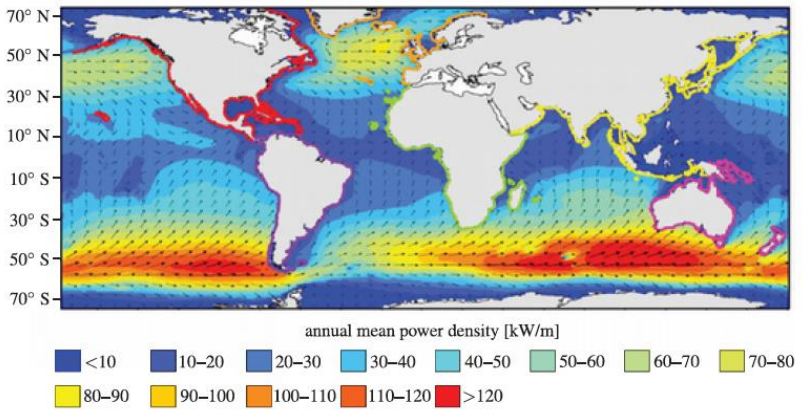
- Not easy to use
  - Distributed over the oceans

  → About 2-3 TW available on coastlines (global need : 20 TW)

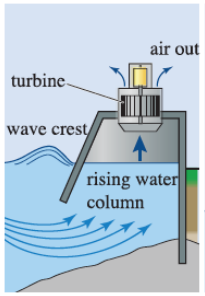
  - Highly variable

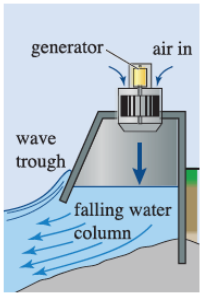
  → requires very robust devices

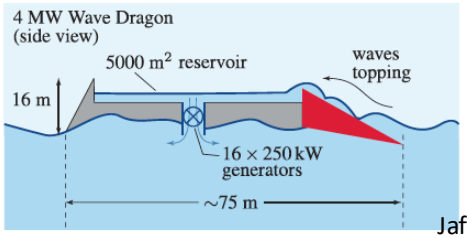
  - Pilots
    - many proposed designs
    - up to 90% efficiency











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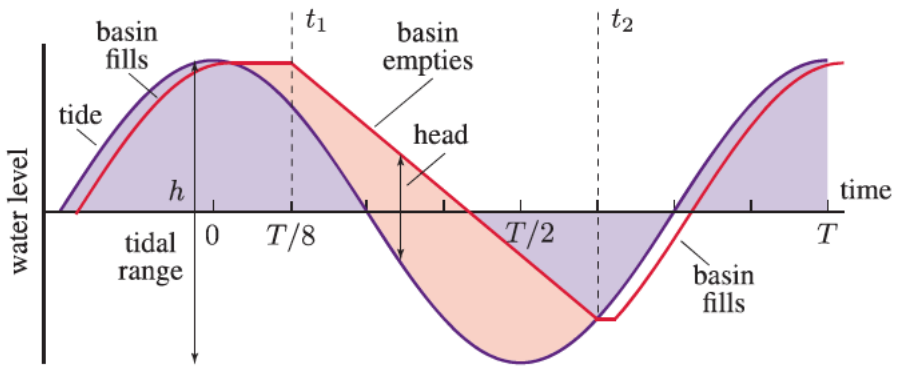
Tidal and current power

- Global tidal power
  - Estimated from increase of Earth-Moon distance: 3.8 cm/yr

  →Increase in Moon angular momentum, decrease in Earth angular momentum

  →Decrease in Earth rotational energy ≈ 10^{20} J/yr ≈ 3 000 GW
  - Mostly lost by friction at seafloor, too diffuse along most coastlines for use
  - Usable only when tides flow in a large reservoir through a narrow channel

  →Estimated potential ~ 80 GW (very limited w.r.t. global need of 20 000 GW)
- tidal barrage : similar to hydropower
- tidal stream turbine : similar to wind turbine







Tidal and current power

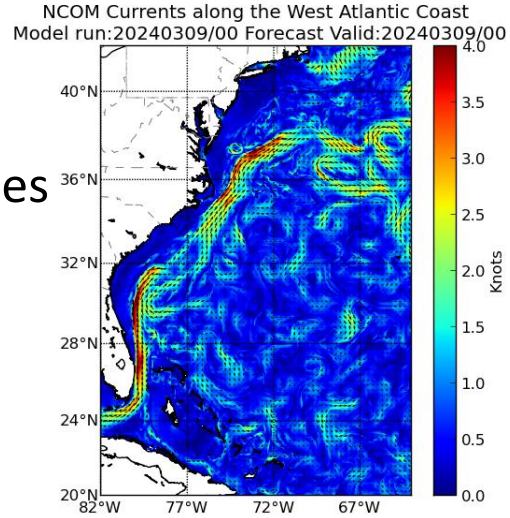
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  - Usable only when tides flow in a large reservoir through a narrow channel

  →Estimated potential ~ 80 GW (very limited w.r.t. global need of 20 000 GW)
- Marine current energy
  - Mostly too slow (< 1 m/s) and too deep to be used
  - Interesting for shallow currents > 2-3 m/s near coastlines

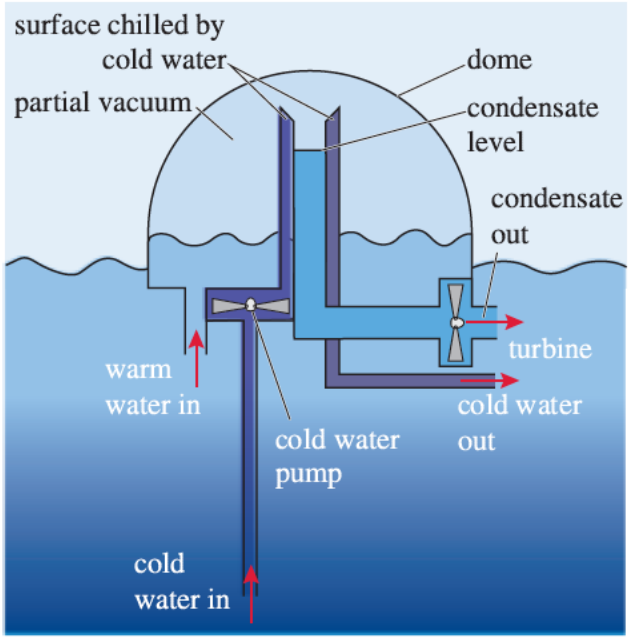
  →Ex: Gulf stream, 250 GW ~ 10-20 GW usable
  - Like tidal energy : very limited w.r.t. global need



National Oceanic and Atmospheric Administration

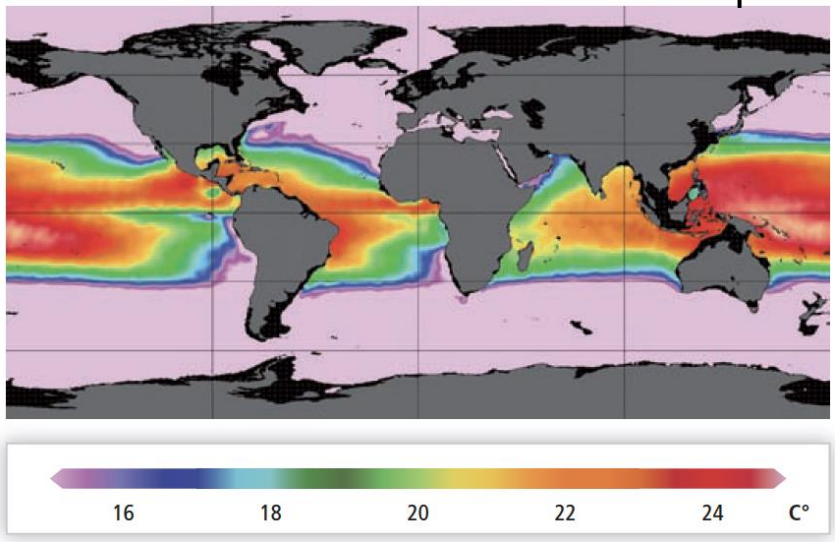
Ocean thermal energy conversion

- Use the temperature difference between surface and deep water
  - Much larger resource than all other forms of ocean energy (thousands of TW)
  - But poor efficiency (Carnot + heat losses)
    - → Minimum of 20°C difference to operate (Carnot η = 6.6%)
    - → Existing pilots : 2-3 %
  - →Global potential : 5 TW



Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press

Average ocean temperature differences between 20 and 1 000 m water depth



IPCC Special Report on Renewable Energy Sources and Climate Change Mitigation (2011)

Key points

- Basics of fluid mechanics:
  - Continuum description, kinetic energy and power density
  - Conservation laws: mass (**continuity equation**), momentum (**Navier-Stokes**), energy (**energy balance equation**)
  - Simplifying assumptions (incompressible, adiabatic, Newtonian / ideal fluid, stationary)
  - **Bernoulli**'s principle
- Working principle of an airfoil:
  - Concepts of **circulation** and **vorticity**
  - **Kutta-Zhukovskii** theorem, d'**Alembert's paradox**
  - General form of drag and lift force: **drag and lift coefficients**, dependence to the angle of attack
- Application to energy:
  - **Betz** limit
  - Horizontal axis wind turbine and **blade design**
  - Hydropower efficiency